
EXERCISES

AI 504 activity: Proving

AI504 — Knowledge Representation

Simon Holm
March, 2026



DEPARTMENT OF MATHEMATICS
AND COMPUTER SCIENCE

Contents

1	How to use $\models$	3
2	Exercise 1.9 from book	3

1 How to use $\models$

- $\underbrace{\Gamma \models \phi}_{\text{entails}} := \forall \mathcal{M}. \mathcal{M} \models \Gamma \rightarrow \mathcal{M} \models \phi$
- $\underbrace{\mathcal{M} \models \phi}_{\text{satisfies}} := \llbracket p \rrbracket \subseteq \llbracket q \rrbracket, i.e. \forall x \in \llbracket p \rrbracket, x \in \llbracket s \rrbracket$
- $\underbrace{\mathcal{M} \models \Gamma}_{\text{satisfies all}} := \forall \phi \in \Gamma. \mathcal{M} \models \phi$

2 Exercise 1.9 from book

See description from book (logic_from_language)

(a) Show that $\mathcal{N} \models \Gamma$

$$H_1 : \Gamma \models p \text{ all } q = \mathcal{N} \models \Gamma \vdash \forall (r \text{ all } s) \in \Gamma. \mathcal{N} \models r \text{ all } s$$

Let $r \text{ all } s \in \Gamma(k)$. We show $\mathcal{N} \models r \text{ all } s, i.e. \llbracket r \rrbracket \subseteq \llbracket s \rrbracket, i.e. \forall x \in \llbracket r \rrbracket, x \in \llbracket s \rrbracket$

$$H_2 : \Gamma \models p \text{ all } q, K = r \text{ all } s \in \Gamma \vdash \forall x \in \llbracket r \rrbracket. x \in \llbracket s \rrbracket$$

Let $x \in \llbracket r \rrbracket, i.e. \llbracket r \rrbracket = \{*\}, i.e. \Gamma \vdash p \text{ all } r, i.e. \exists \mathcal{D} \in \text{Der}_{\Gamma, p \text{ all } r}(L)$.

We show $x \in \llbracket s \rrbracket, i.e. \llbracket s \rrbracket = \{*\}, i.e. \Gamma \vdash p \text{ all } s, i.e. \exists \mathcal{D}' \in \text{Der}_{\Gamma, q \text{ all } s}$

$$H_3 : \Gamma \models p \text{ all } q, K : r \text{ all } s \in \Gamma, L : \exists \mathcal{D} \in \text{Der}_{\Gamma, p \text{ all } r} \vdash \exists \mathcal{D}' \in \text{Der}_{\Gamma, p \text{ all } s}$$

We choose $\mathcal{D}' := \frac{\text{all } p \text{ are } r \quad \text{all } r \text{ are } s}{\text{all } p \text{ are } s} \text{ BARBARA.}$

$$\frac{\text{all } p \text{ are } r \quad \text{all } r \text{ are } s}{\text{all } p \text{ are } s} \text{ BARBARA.} \in \text{Der}_{\Gamma, p \text{ all } s},$$

$i.e. p \text{ all } r \in \text{Der}_{\Gamma, p \text{ all } r}$ and $r \text{ all } s \in \text{Der}_{\Gamma, r \text{ all } s}$.

Obvious by K and L .

(b) Show that $\mathcal{N} \models p \text{ all } q$ [Hint: Use part (1).]

We want to show that $\mathcal{N} \models \Gamma \Rightarrow \mathcal{N} \models p \text{ all } q$

$$H_4 : \Gamma \models p \text{ all } q \vdash \mathcal{N} \models \Gamma \Rightarrow \mathcal{N} \models p \text{ all } q$$

To prove $\mathcal{N} \models \Gamma \rightarrow \mathcal{N} \models p \text{ all } q$, we show that $\mathcal{N} \in \text{Models}$ and $\forall \mathcal{M} \in \text{Models}$.

$\mathcal{N} \models \Gamma \rightarrow \mathcal{N} \models p \text{ all } q$. Obvious by H

(c) Use part (2) to show that $\Gamma \vdash p$ all q

$$\begin{aligned} H_5 : \Gamma \models p \text{ all } q \vdash \mathcal{N} \vdash p \text{ all } q \\ = \Gamma \models p \text{ all } q \vdash \llbracket q \rrbracket = \{*\} \end{aligned}$$

We prove that $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket \wedge \llbracket p \rrbracket = \{*\}$:

- $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$, we know this by definition of satisfiability
- $\llbracket p \rrbracket = \{*\}$, *i.e.* $\Gamma \vdash p$ all p This is obvious by AXIOM.